Week 11 — TikZ Diagrams

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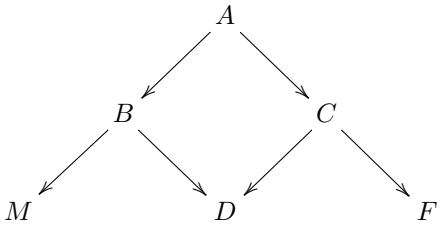
November 2, 2026

1 XYPic diagrams

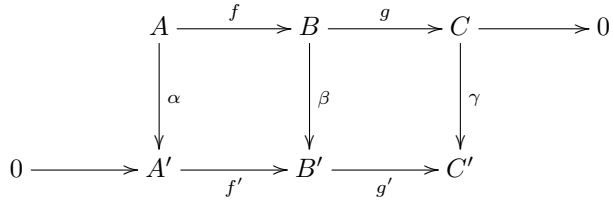
These are some gaussian elimination steps from my notes converted to xypic.

$$\begin{bmatrix} 1 & 2 & 4 & -4 \\ 5 & 11 & -21 & -22 \\ 3 & -2 & 3 & 11 \end{bmatrix} \xrightarrow[R_3 \leftarrow R_3 - 3R_1]{R_2 \leftarrow R_2 - 5R_1} \begin{bmatrix} 1 & 2 & 4 & -4 \\ 0 & 1 & -41 & -2 \\ 0 & -8 & -9 & 23 \end{bmatrix} \xrightarrow{R_3 \leftarrow 8R_2 + R_3} \begin{bmatrix} 1 & 2 & 4 & -4 \\ 0 & 1 & -41 & -2 \\ 0 & 0 & -337 & 7 \end{bmatrix}$$

I attempted to do a binary tree in xypic. Not the prettiest way to code a tree, but it outputs...

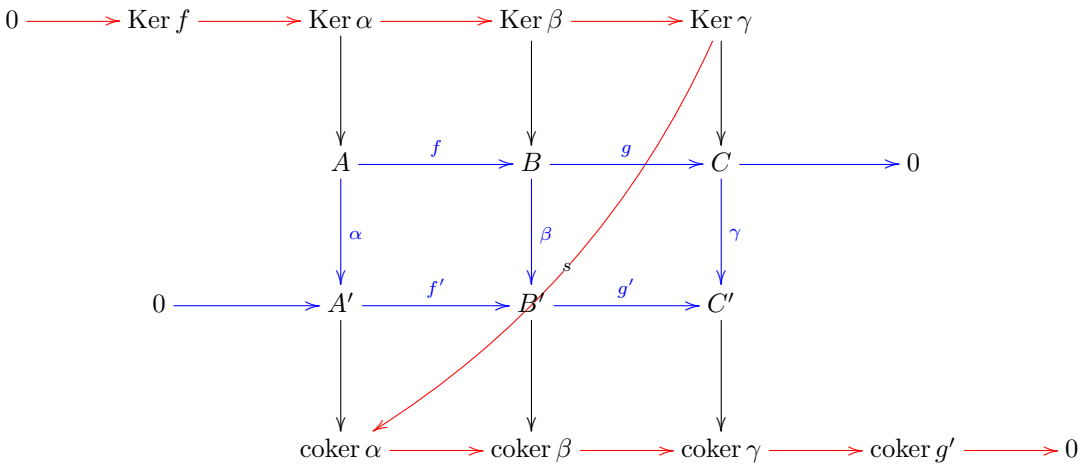


2 Snake Lemma 1



3 Snake Lemma 2

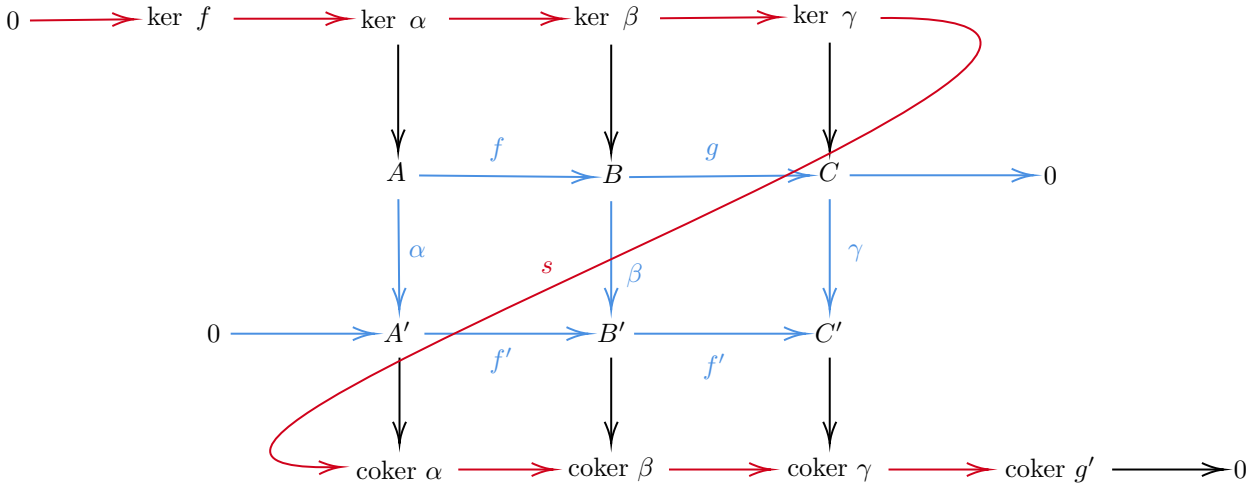
is there a way to say "every right arrow on this line should be blue"?



I do not think the snake lemma drawing is possible in xypic with that entry/exit curve, at least I was not able to figure out how to do it.

4 Mathcha and TikZ

I am here trying to do it in tikz, exported from Mathcha. I used a spline to create the curved arrow, which I could not figure out in xypic.



5 Note

AI was not able to figure out how to do the curved arrow in xypic at all. Google search AI (gemini) was not able to either, but recommended using `\crv` but was not useful.